

Approximate Belief-Selective Propagation Detector for Massive MIMO Systems

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Abstract—When faced with challenging antenna configurations or high-order modulations in realistic propagation environments, the Belief Propagation (BP) MIMO detector outperforms its linear counterparts. To mitigate the error floor issue and lower the complexity, a revised BP detector, named the Belief-selective Propagation (BsP) detector, has recently emerged by selectively utilizing trusted incoming messages for updates. Despite those promising potentials, the straightforward hardware implementation of the BsP detector still suffers from high complexity and necessitates further optimization. To bridge the gap between the BsP algorithm and implementation, this paper introduces an *approximate* but implementation-friendly BsP detector called aBsP, based on which the very first BsP hardware is proposed. Two unexplored features: *approximate initialization* and *simplified message updates* save the complexity (more than 84%) with acceptable performance penalization. Multi-level optimization techniques involving group-layered message updating, approximate arithmetic circuits, and hybrid-precise quantization are developed to boost the hardware efficiency. A 128×8 256-QAM aBsP MIMO detector ASIC in 40 nm CMOS occupies an area of 0.68 mm^2 and reaches a throughput of 790.52 Mbps. Benchmarking with the recent arts, this work achieves $1.08\times$ area efficiency and $3.34\times$ gate efficiency.

Index Terms—MIMO detector, BsP detector, approximate BsP (aBsP), hardware implementation, ASIC.

I. INTRODUCTION

THANKS to its significant improvements in spectral efficiency and channel capacity, massive multiple-input multiple-output (MIMO) has emerged as a fundamental pillar in the realm of beyond 5G (B5G) and the forthcoming

6G era [1], [2]. However, as the scale of antennas continues to expand, designing a MIMO detector that balances performance and implementation has become increasingly challenging. As the most common MIMO detector, the linear minimum mean square error (LMMSE) detector [3] is widely adopted. However, in B5G/6G scenarios, e.g., enhanced mobile broadband (eMBB) and ultra-reliable low-latency communication (uRLLC), the demanding antenna configurations and high-order modulations exacerbate the performance degradation of the LMMSE detector. On the other hand, message-passing detectors [4], [5], [6], [7], [8], [9], [10], [11] exhibit superior performance over the LMMSE one, with the Expectation Propagation (EP) detector [5], [6], [9], [10], [11] standing out as a notable example. However, the EP detector inherently suffers from high computational complexity attributed to the iterative non-linear operations, hindering its efficient implementation.

Alternatively, Belief Propagation (BP) detector [12], [13], [14], whose core operations are multiplications (Mul.) and additions (Add.), enjoys superior performance to its linear counterparts in challenging antenna configurations or high-order modulations. Benefiting from its fully parallel soft-in soft-out message-updating architecture, the BP detector exhibits two merits: 1) high data throughput, and 2) great potential in coded systems. Moreover, the BP algorithm offers one promising framework for an all-message-passing unified baseband processor [15]. However, the original BP detector [12] suffers from exponential costs as the MIMO scale increases. Therefore, the *Gaussian* approximation of interference BP (GAI-BP) [13] and its real-domain extension (called RD-GAI-BP) [14] are proposed, which lower the complexity but confront severe performance error floor issue. To address this issue, the Belief-selective Propagation (BsP) detector has been recently proposed in [16] and draws intensive attention. By exploring the sparsity of the “beliefs”, the BsP detector only exploits the incoming messages with high reliability for updates, thus not only relieving the performance error floor but also reducing computational costs.

To verify the feasibility of BsP and showcase its advantages, it is imperative to develop its hardware architecture with ASIC implementation. However, the straightforward mapping of the BsP algorithm is intractable due to its LMMSE initialization and the full-edge connections. To bridge the gap between the

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BsP algorithm and hardware, an implementation-friendly version of the BsP, named approximate BsP (aBsP), is proposed. And the first BsP ASIC is presented. The main contributions of this work are listed as follows.

- To reduce the complexity of the BsP, the aBsP detector is proposed. Two unexplored features: *approximate initialization* and *simplified message updates* are developed, achieving more than 84% complexity reduction. Specifically, a weighted *Neumann* series approximation-based MMSE (wNSA-MMSE) is utilized to circumvent the matrix-inversion operations in initialization. Message elimination and lattice point reduction are employed to reduce the message-updating complexity. Numerical results demonstrate that the aBsP detector achieves superior performance over its counterparts in B5G/6G scenarios.
- To boost the hardware efficiency of the aBsP, multi-level area/power optimizations are introduced. In particular, first, a group-layered message-updating scheme is designed to lower the latency. Second, approximate arithmetic circuits corresponding to the multiplication and division operations are explored to shorten the critical path. Finally, a hybrid-precise quantization scheme is provided to compress the average bit-width of aBsP detector, thus relieving the hardware costs.
- To demonstrate the hardware feasibility, a 0.68 mm² 790.52 Mbps aBsP ASIC at 437.50 mW power is implemented in 40 nm COMS for 128 × 8 256-QAM MIMO systems. Two main arithmetic modules: 1) wNSA-MMSE process element (PE) and 2) aBsP Message-updating PE are designed to map the aBsP algorithm to its hardware architecture. ASIC implementation results exhibit that benchmarking with the recent arts, the aBsP detector succeeds in achieving 1.08× area efficiency (AE) and 3.34× gate efficiency (GE).

The remainder of this paper is organized as follows. Section II introduces the basics of the BsP detector and the other preliminaries. Section III proposes the aBsP detector and provides its performance evaluation together with complexity analysis. Section IV develops the multi-level area/power optimization techniques. ASIC result comparisons of the aBsP and the recent arts are provided in Section V. Finally, Section VI summarizes the entire paper.

II. PRELIMINARIES

A. System Model

Consider an uplink massive multi-user MIMO (MU-MIMO) system involving a gNodeB (gNB) equipping with N_r (usually hundreds or thousands) antennas and N_t single-antenna user equipments (UEs). In an uplink time slot, N_t UEs jointly transmit an i.i.d. complex vector $\tilde{\mathbf{x}} = [\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_{N_t}]^T$ and the gNB receives the signal $\tilde{\mathbf{y}} = [\tilde{y}_1, \tilde{y}_2, \dots, \tilde{y}_{N_r}]^T$ as

$$\tilde{\mathbf{y}} = \tilde{\mathbf{H}}\tilde{\mathbf{x}} + \tilde{\mathbf{n}}, \quad (1)$$

where $\tilde{\mathbf{H}} = [\tilde{\mathbf{h}}_1, \tilde{\mathbf{h}}_2, \dots, \tilde{\mathbf{h}}_{N_r}]^T$ is a channel matrix involving $N_r \times N_t$ complex channel coefficients, and $\tilde{\mathbf{n}}$ is the noise

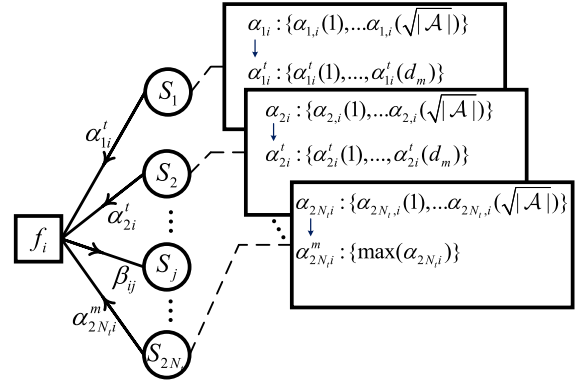


Fig. 1. Factor graph of the BsP detector with single FN and $2N_t$ SNs.

following the complex *Gaussian* distribution $\mathcal{CN}(0, \sigma_n^2)$. Let $\tilde{\mathbf{h}}_i$ denote the i -th row vector of $\tilde{\mathbf{H}}$, and then each coefficient h_{ij} follows the complex *Gaussian* distribution as $h_{ij} \sim \mathcal{CN}(0, 1)$. Define M as the order of quadrature amplitude modulation (QAM) for each symbol. Then the cardinality of the constellation set is denoted as $|\mathcal{A}| = 2^M$. In the message-passing based MIMO detection algorithm, the real-domain expression of (1) is often used, written as

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}, \quad (2)$$

where $\mathbf{y} = \begin{bmatrix} \Re(\tilde{\mathbf{y}}) \\ \Im(\tilde{\mathbf{y}}) \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} \Re(\tilde{\mathbf{x}}) \\ \Im(\tilde{\mathbf{x}}) \end{bmatrix}$, $\mathbf{n} = \begin{bmatrix} \Re(\tilde{\mathbf{n}}) \\ \Im(\tilde{\mathbf{n}}) \end{bmatrix}$, $\mathbf{H}$ is acquired by

$$\mathbf{H} = \begin{bmatrix} \Re(\tilde{\mathbf{H}}) & -\Im(\tilde{\mathbf{H}}) \\ \Im(\tilde{\mathbf{H}}) & \Re(\tilde{\mathbf{H}}) \end{bmatrix}, \quad (3)$$

and $\Re(*)$ together with $\Im(*)$ represents the real part and the imaginary part of $*$, respectively. It is noted that $\mathbf{y}$ and $\mathbf{n}$ hold the dimension of $2N_r \times 1$, $\mathbf{x}$ has the dimension of $2N_t \times 1$, and $\mathbf{H}$ involves the dimension of $2N_r \times 2N_t$. In addition, $\mathbf{x}$ consists of the real-domain symbols which are chosen from the real constellation with cardinality of $\sqrt{|\mathcal{A}|}$.

B. BsP Detector

As represented in [16], the BsP detector only selects the lattice points with large reliability for message updates. Fig. 1 shows the factor graph of the BsP detector with a single factor node (FN) and $2N_t$ symbol nodes (SNs). The message β_{ij} sent from the i -th FN to the j -th SN is computed by (4), as shown at the bottom of the next page, where $\mathcal{B}(d_m, d_f)$ is the configuration set regarding d_f chosen edges and $2N_t - d_f - 1$ simplified edges. For each chosen edge, d_m lattice points with relatively large reliability are searched indicated by the incoming α message. In addition, for each simplified edge, only the lattice point with the largest reliability is utilized to compute the β_{ij} . The configuration set $\mathcal{B}(d_m, d_f)$ is illustrated as

$$\mathcal{B}(d_m, d_f) = \left\{ \mathbf{s}_{\mathbf{k}/j} = \begin{bmatrix} \mathbf{s}_{\mathbf{k}/j, d_f}^T & \bar{\mathbf{s}}_{\mathbf{k}/j, d_f}^T \end{bmatrix}^T \right\}, \quad (5)$$

where $\mathbf{s}_{\mathbf{k}/j}$ denotes the possible lattice point vector without symbol s_j , $\mathbf{s}_{\mathbf{k}/j, d_f}$ and $\bar{\mathbf{s}}_{\mathbf{k}/j, d_f}$ represent the lattice point

vector corresponding to the chosen edges and simplified edges, respectively. Alongside the β message updates, the α messages are computed by

$$\alpha_{ji}(k) = \sum_{t=1, t \neq i}^{2N_r} \beta_{tj}(k). \quad (6)$$

When the BsP detector achieves the maximum number of iterations, it will stop message propagation and output the soft information for each transmitted symbol as

$$\gamma_j(k) = \sum_{i=1}^{2N_r} \beta_{ij}(k). \quad (7)$$

The BsP detector relies on an LMMSE pre-processor to provide the initial information for each transmitted symbol. In particular, denote $\hat{s}_{\text{init}_j}$ as the mean of the transmitted symbol s_j computed by the pre-processing detector, and then the initial symbol *a priori* probability can be computed by

$$p_j(\mu_k) = \exp \left\{ -\frac{\|\mu_k - \hat{s}_{\text{init}_j}\|^2}{2\sigma_{\text{init}_j}^2} \right\}, \quad (8)$$

where $\sigma_{\text{init}_j}^2$ is the j -th diagonal element of the noise covariance matrix $\mathbf{K} = (\mathbf{H}^H \mathbf{H} + \sigma^2 \mathbf{I})^{-1}$. With the initial symbol *a priori* probability, the γ messages computed by

$$\gamma_j(k) = \log \frac{p_j(\mu_k)}{p_j(\mu_1)} \quad (9)$$

are utilized to activate the iterations of BsP detection. For better understanding, we summarize the BsP detection algorithm in Alg. 1. It is noted that without special explanation, only configuration $\mathcal{B}(1, 1)$ is considered in this work. More details of the BsP detector can be found in [16].

III. APPROXIMATE ALGORITHM FOR HARDWARE IMPLEMENTATION

The aBsP detection algorithm is proposed in this section to support the efficient hardware implementation. To this end, the wNSA-MMSE detector is first adopted to initialize the BsP message updates, and then two algorithmic approximation techniques for message updates are employed to simplify the computations. Finally, complexity analysis together with the performance evaluation is provided to demonstrate the effectiveness of the proposed aBsP detector.

A. Approximate Initialization

Hardware-friendly pre-processor is required to provide the initial *a priori* information for the aBsP detector. The wNSA technique [17] can iteratively approximate the convergent results of the MMSE detection while bypassing the matrix-inversion operations. Therefore, we adopt the wNSA technique

Algorithm 1 BsP Detection Algorithm

Input : $\mathbf{y}, \mathbf{H}, \mathcal{A}, Q_L, d_m, d_f$
Output: symbol soft message γ

```

1 initialization:  $\hat{\mathbf{s}}_{\text{init}} \leftarrow \text{LMMSE}(\mathbf{y}, \mathbf{H})$ ;
2 for  $j \leftarrow 1$  to  $2N_t$  do // compute  $\gamma$ 
3   Derive  $\gamma_j$  from (8), (9);
4   for  $i \leftarrow 1$  to  $2N_r$  do
5      $\alpha_{ji} = \gamma_j$ ;
6 for Iter.  $\leftarrow 1$  to  $Q_L$  do
7   for  $i \leftarrow 1$  to  $2N_r$  do // update  $\beta$ 
8     for  $j \leftarrow 1$  to  $2N_t$  do
9       find the maximum value of  $\alpha_{ji}$ ;
10      build  $\mathcal{B}(1, 1)$  as (5);
11      update  $\beta_{ij}$  as (4);
12   for  $j \leftarrow 1$  to  $2N_t$  do // compute  $\alpha$ 
13     update  $\gamma_j$  as (7);
14     for  $i \leftarrow 1$  to  $2N_r$  do
15       update  $\alpha$  with (6);
16 output  $\gamma$  messages.
```

to initially estimate each transmitted symbol, and then further simplify the computation of the initial γ messages based on the wNSA-MMSE results. In particular, suppose that $\mathbf{W} = \sigma_n^{-2} \mathbf{H}^T \mathbf{H} + \Lambda$, where $\Lambda = E_s^{-1} \mathbf{I}$, and $\mathbf{D}$ consists of main diagonal elements of $\mathbf{W}$. The wNSA results of K -term is computed by

$$\mathbf{W}^{-1} \approx \sum_{n=0}^{K-1} (\mathbf{I} - (\omega \mathbf{D}^{-1}) \mathbf{W})^n (\omega \mathbf{D}^{-1}), \quad (10)$$

where ω is the weighted parameter used to accelerate convergence [17]. The wNSA-MMSE pre-processor iteratively updates the estimation of each transmitted symbol, until it reaches the maximum number of iterations, ultimately providing estimates, denoted as $\hat{\mathbf{s}}_{\text{init}}$, for each lattice point. The $\hat{\mathbf{s}}_{\text{init}}$ is then utilized to compute the initial γ messages according to (8) and (9). However, this computation brings about a substantial hardware resource overhead, involving additional $2N_t \sqrt{|\mathcal{A}|}$ Mul., Add., and exponentiations (Exp.) (each FN receives the same initial messages). To address this issue, we simplify the computation of the initial γ messages by calculating the distance between the estimates and the lattice points, which can greatly diminish redundant calculations. Specifically, with the estimates $\hat{\mathbf{s}}_{\text{init}}$, γ is computed by

$$\gamma_j(k) = -\text{abs}(\mu_k - \hat{s}_{\text{init}_j}), \quad (11)$$

Utilizing (11), the computation of the initial γ only necessitates $2N_t \sqrt{|\mathcal{A}|}$ additions.

$$\beta_{ij}(k) = \max_{\substack{s: s_j = \mu_k, \\ \mathbf{s}_{\mathbf{k}/j} \in \mathcal{B}(d_m, d_f)}} \left\{ -\frac{1}{2\sigma^2} \|\mathbf{y}_i - \mathbf{h}_i \mathbf{s}\|^2 + \sum_{t=1, t \neq j}^{2N_t} \alpha_{ti} \right\} - \max_{\substack{s: s_j = \mu_1, \\ \mathbf{s}_{\mathbf{k}/j} \in \mathcal{B}(d_m, d_f)}} \left\{ -\frac{1}{2\sigma^2} \|\mathbf{y}_i - \mathbf{h}_i \mathbf{s}\|^2 + \sum_{t=1, t \neq j}^{2N_t} \alpha_{ti} \right\}. \quad (4)$$

Remark 1: Note that the weighted Gram matrix $\mathbf{A} = \sigma_n^{-2} \mathbf{H}^T \mathbf{H}$ and weighted matching filter $\mathbf{b} = \sigma_n^{-2} \mathbf{H}^T \mathbf{y}$ remain constant over the channel coherence interval [6], which can be pre-calculated before and re-used in each iteration of the aBsP detector for high efficiency.

B. Simplified Message Updates

To reduce hardware implementation complexity and enhance hardware feasibility of the BsP, we further develop two key approximation techniques, which involve message elimination and lattice point reduction, for the message updating scheme. Details of the proposed approximation techniques are provided below.

1) *Message elimination:* The calculation of α messages costs $4N_r N_t |\sqrt{A}|$ Add. per iteration, reducing the efficiency of hardware implementation. In fact, the message updating of the BsP could bypass α message computation if the configuration set $\mathcal{B}(1, 1)$ is employed. With $\mathcal{B}(1, 1)$, the β message updating in (4) is simplified as

$$\beta_{ij}(k) = \frac{(y_i - \phi_{ij} - h_{ij}\mu_1)^2}{2\sigma_n^2} - \frac{(y_i - \phi_{ij} - h_{ij}\mu_k)^2}{2\sigma_n^2}. \quad (12)$$

where $\phi_{ij} = \mathbf{h}_{i/j} \mathbf{s}_{k/j}$, and the $\mathbf{s}_{k/j}$ is directly determined by the associated γ messages. Therefore, the β messages can be derived without the α messages in this case. Based on this observation, we circumvent the α messages in our aBsP detector and only update the γ messages in each iteration.

2) *Lattice point reduction:* We employ the lattice point reduction technique to redevelop the configuration set of each β message for further complexity reduction. Specifically, denote $\mathcal{C}_j(n_m)$ as the configuration set of the chosen lattice points corresponding to the j -th transmitted symbol estimated by wNSA-MMSE. Then $\mathcal{C}_j(n_m)$ is written as:

$$\mathcal{C}_j(n_m) = \{\mu_\psi = [\mu_{\psi_1}, \mu_{\psi_2}, \dots, \mu_{\psi_{n_m}}] : \forall \psi, \gamma_j(\psi_1) > \gamma_j(\psi_2) > \dots > \gamma_j(\psi_{n_m})\}, \quad (13)$$

where $\{\psi_1, \psi_2, \dots, \psi_{n_m}\}$ are the indices of constellation points corresponding to the n_m largest γ_j value, with γ_j is derived from the wNSA-MMSE pre-processor. Therefore, we first build up the configuration set of reduced lattice points for each transmitted symbol s_j with (13) after the wNSA-MMSE initialization. Then, the message updating of each β_{ij} only relies on the elements within $\mathcal{C}_j(n_m)$. To ensure completeness, we set the un-updated elements of β_{ij} to zeros. Finally, the β messages in the aBsP detector are constructed as

$$\beta_{ij} = \{[\beta_{ij}(\psi_1), \dots, \beta_{ij}(\psi_{n_m}), \mathbf{0}] \mid \mu_\psi \in \mathcal{C}_j(n_m)\}. \quad (14)$$

It is noted that we only construct each $\mathcal{C}_j(n_m)$ once before the iterations of the aBsP, and it will not be updated thereafter. By utilizing the algorithmic approximations previously described, the proposed aBsP achieves a notable reduction in computational complexity and enjoys better hardware implementation. Considering the detailed arithmetic, the proposed aBsP is summarized as Alg. 2.

TABLE I
MAXIMUM NUMBER OF ITERATIONS FOR DIFFERENT DETECTORS

Detector	Iteration (Rayleigh, WINNER-II) channel		
	$N_t = 8$	$N_t = 32$	$N_t = 64$
EPD [6]	(5, 5)	(5, 5)	(5, 5)
LAMA [7]	(20, 30)	(20, 30)	(20, 30)
ILMPD [10]	(4, 4)	(7, 10)	(40, 50)
EPA-SORI [11]	(3, 10)	(3, 30)	(7, 40)
BsP, $\mathcal{B}(1, 1)$ [16]	(2, 2)	(5, 5)	(7, 10)
Proposed aBsP	(2, 2)	(5, 5)	(10, 10)

Algorithm 2 aBsP Detection Algorithm

Input : $\mathbf{y}, \mathbf{H}, \mathcal{A}, Q_L, n_m$
Output: symbol soft message γ

```

1 initialization:  $\hat{\mathbf{s}}_{\text{init}} \leftarrow \text{wNSA}(\mathbf{y}, \mathbf{H})$ ;
2 for  $j \leftarrow 1$  to  $2N_t$  do // compute  $\gamma$ 
3    $\gamma_j = -\text{abs}(\sqrt{A} - \hat{\mathbf{s}}_{\text{init},j})$ ;
4   configure truncated lattice set
    $\mathcal{C}_j(n_m)$  as (13);
5 for Iter.  $\leftarrow 1$  to  $Q_L$  do
6   for  $i \leftarrow 1$  to  $2N_r$  do // update  $\beta$ 
7      $R_{\text{inc}} = \sum_{j=1}^{2N_t} h_{ij} \mu_{\psi_j}$ ;
8     for  $j \leftarrow 1$  to  $2N_t$  do
9        $\mathbf{R} = R_{\text{inc}} - h_{ij} \sqrt{A} (\mathcal{C}_j(n_m))$ ;
10       $\tilde{\mathbf{y}} = \mathbf{y}_i - \mathbf{R}$ ;  $\tilde{\mathbf{y}}_2 = 2h_{ij} \tilde{\mathbf{y}}$ ;
11       $\mathbf{F}_1 = \sqrt{A} (\mathcal{C}_j(n_m)) - \mu_1$ ;
12       $\mathbf{F}_2 = \mu_1^2 - \sqrt{A} (\mathcal{C}_j(n_m)) \times \sqrt{A} (\mathcal{C}_j(n_m))$ ;
13       $\tilde{\mathbf{y}}_t = \tilde{\mathbf{y}}_2 \times \mathbf{F}_1$ ;  $\mathbf{H}_u = h_{ij}^2 \mathbf{F}_2$ ;
14       $\beta_{ij} = \mathbf{0}$  // reset the  $\beta$ ;
15       $\beta_{ij}(\mathcal{C}_j(n_m)) = \tilde{\mathbf{y}}_t + \mathbf{H}_u$ ;
16   for  $j \leftarrow 1$  to  $2N_t$  do // update  $\gamma$ 
17     for  $k \leftarrow 1$  to  $n_m$  do
18        $\gamma_j(\mathcal{C}_j(k)) = \sum_{i=1}^{N_r} \beta_{ij}(\mathcal{C}_j(k))$ ;
19     update  $\psi_j$  via  $\gamma_j$ -based tournament;
20 output  $\gamma$  message.
```

C. Performance Evaluation

To demonstrate the performance of the proposed aBsP, we compare the uncoded BER results of the aBsP detector with that of the LMMSE [3], EPD [6], LAMA [7], ILMPD [10], EPA-SORI [11], and previous BsP with $\mathcal{B}(1, 1)$ [16] in uplink $128 \times \{8, 32, 64\}$ massive MIMO systems with 256-QAM modulations. Two typical MIMO channel models are employed, involving Rayleigh channel and WINNER-II channel [18], to see the robustness of the proposed aBsP in realistic MIMO scenarios. The maximum number of iterations (denoted as Q_L) employed for different MIMO scenarios with respect to the proposed aBsP and the compared detectors are exhibited in Table I. Note that the maximum number of the iterations of the ILMPD and EPA-SORI are employed following the parameters in [10] and [11], given the same MIMO scenarios. However, for the more “severe” MIMO scenarios, e.g., MIMO

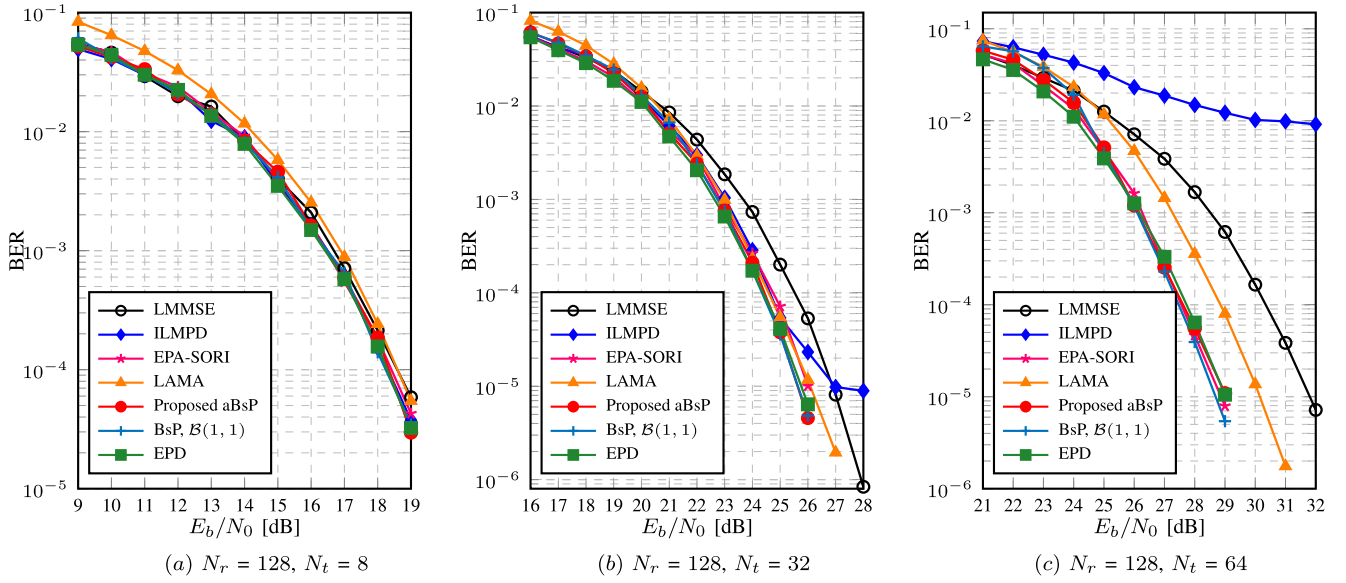


Fig. 2. Unencoded BER of the $128 \times \{8, 32, 64\}$ uplink MIMO detections with 256-QAM modulations in Rayleigh channel. The maximum number of iterations are shown as Table I.

scenarios supporting more UE or MIMO scenarios under correlated channels, we scale up the maximum number of iterations for the ILMPD and EPA-SORI to show their best error performance. For the proposed aBsP, $Q_L = \{2, 5, 10\}$ is set for the MIMO scenarios with $N_t = \{8, 32, 64\}$. Denote K as the maximum number of iterations for the pre-processors. For the EPA-SORI, the same K as that in [11] with Rayleigh channel and $K = \{10, 30, 50\}$ for the $128 \times \{8, 32, 64\}$ MIMO scenarios with WINNER-II channel are set up for simulation. For the proposed aBsP, we set $K = \{7, 10, 15\}$ for the $128 \times \{8, 32, 64\}$ MIMO scenarios with Rayleigh channel and $K = \{10, 15, 20\}$ with WINNER-II channel. In addition, $n_m = 2$ is selected to enable the lattice point reduction technique for the proposed aBsP detector.

Fig. 2 reports the performance comparison of the proposed aBsP detector and other detectors under Rayleigh channels. In the 128×8 MIMO scenario, the proposed aBsP converges to the consistent results with that of the LMMSE, ILMPD, EPA-SORI, LAMA, and EPD. However, performance advantages are displayed for the MIMO systems supporting a larger number of UE than state-of-the-art detectors. Specifically, compared with the LMMSE, the proposed aBsP detector achieves 1.0 and 2.8 dB SNR gains at $\text{BER} = 10^{-4}$ in the MIMO scenarios with $N_t = 32$ and $N_t = 64$, respectively. Compared with the LAMA, the near-optimal detector also shows slight performance gains in MIMO scenarios with $N_t = 32$ and earns more than 1.2 dB SNR gains in 128×64 MIMO scenarios. The ILMPD suffers from error floor in relatively high-SNR regions in the 128×32 MIMO scenario, and the error floor is dramatically exacerbated in the 128×64 MIMO scenario, thus causing system failure. Therefore, the proposed aBsP detection demonstrates notable superiority in robustness compared to the ILMPD. As the approximate version of the BsP detector, the proposed aBsP detector remains a tight BER performance compared with the BsP, $\mathcal{B}(1, 1)$ in variable MIMO scenarios.

Fig. 3 illustrates the comparisons among the proposed aBsP and other detectors with the same antenna configurations as Fig. 2, but under WINNER-II channels [18]. A uniform circular array (UCA) of 128 elements with 1 m radius is deployed in the gNB and N_t UEs are randomly distributed around the gNB, where each UE is equipped with a single transceiver component. The urban macro-cell scenarios together with the non-line-of-sight (NLOS) propagation condition are employed. Fig. 3 remains the same trend as in Fig. 2. However, the EPA-SORI detection suffers from severe error floor in $128 \times \{32, 64\}$ MIMO scenarios under WINNER-II channels. Fig. 4 further shows BER comparison results of the proposed aBsP detector and the other detectors in a 128×64 MIMO system with $\{16, 64, 256\}$ -QAM modulations under Rayleigh channels. The maximum number of iterations for each detector is set the same as that for $N_t = 64$ in Table I. It is observed that the proposed aBsP detector remains a stable near-optimal BER performance in MIMO scenarios equipped with different modulations. Overall, the proposed aBsP detector demonstrates robust BER performance when faced with challenging channels and different modulation schemes.

D. Complexity Analysis

To demonstrate the computational complexity reduced by the algorithmic approximations, we provide the complexity comparison results of the proposed aBsP detector, the BsP detector [16], and the RD-GAT-BP detector [14] here. Following the method in [16], we utilize the number of Mul., Add., and Exp. to roughly measure the computational complexity.

Since the proposed aBsP detector is an approximate version of the BsP detector, we start by analyzing the complexity of the BsP detector with the configuration set $\mathcal{B}(d_m, d_f)$. For each iteration, the BsP detector requires $4N_r N_t (2N_r - 1) d_m^{d_f - 1} \sqrt{|\mathcal{A}|}$ Mul. and Add. to compute β

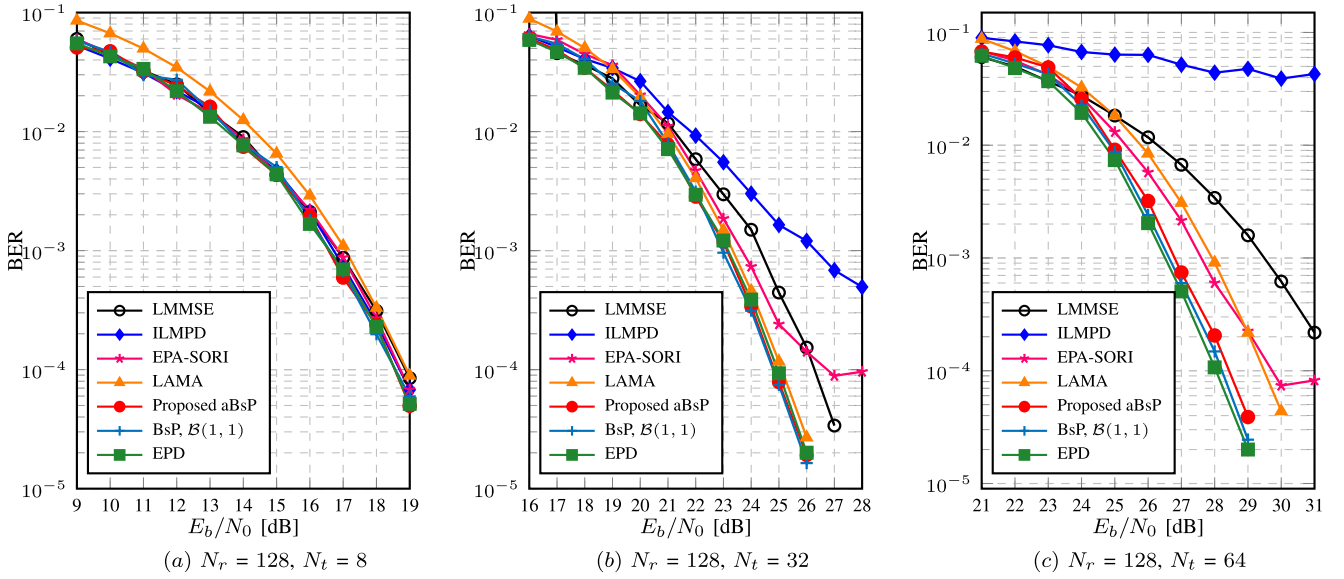


Fig. 3. Uncoded BER of the $128 \times \{8, 32, 64\}$ uplink MIMO detections with 256-QAM modulations in WINNER II channel. The maximum number of iterations is shown in Table I.

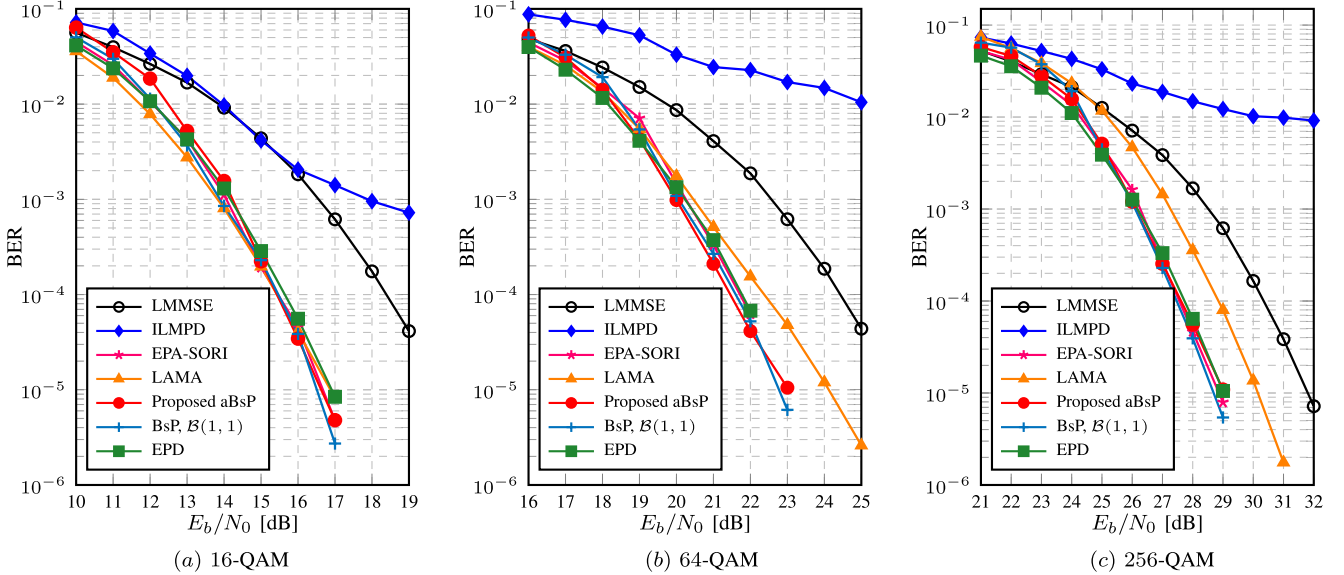


Fig. 4. BER performance comparisons of the proposed aBsP detector and the other detectors in 128×64 MIMO systems with $\{16, 64, 256\}$ -QAM modulations under Rayleigh channels.

messages, and $4N_r N_t \binom{2N_t-1}{d_{f-1}^{d_f-1}} d_m^{d_f-1} \sqrt{|\mathcal{A}|}$ addition to compute α messages. Therefore, with Q_L iterations, the BsP detector involves $4Q_L N_r N_t \binom{2N_t-1}{d_{f-1}^{d_f-1}} d_m^{d_f-1} \sqrt{|\mathcal{A}|}$ Mul. and $8Q_L N_r N_t \binom{2N_t-1}{d_{f-1}^{d_f-1}} d_m^{d_f-1} \sqrt{|\mathcal{A}|}$ Add.. When equipped with the $\mathcal{B}(1, 1)$, this complexity involves $4Q_L N_r N_t \sqrt{|\mathcal{A}|}$ Mul. and $8Q_L N_r N_t \sqrt{|\mathcal{A}|}$ Add.. Here note that this complexity result is an approximate operation number per channel use which does not contain the complexity for initialization. As the simplified version of the BsP, the proposed aBsP detector in this work employs the lattice point reduction (see section III-B) strategy to reduce the computational complexity. Benefiting from this strategy, the proposed aBsP detector only needs to compute n_m lattice points instead of all of the $\sqrt{|\mathcal{A}|}$ lattice points in each iteration. Therefore, its computational complexity involves $4N_r N_t n_m$ Mul. and $8N_r N_t n_m$ Add. for each iteration. As for the RD-GAI-BP detector, we follow the complexity results

in [14], which requires $4N_r N_t \sqrt{|\mathcal{A}|}$ Mul., $8N_r N_t \sqrt{|\mathcal{A}|}$ Add., and $4N_r N_t \sqrt{|\mathcal{A}|}$ Exp. for each iteration. Table II summarizes the complexity results of each detector above, where the complexity results for initialization are also included.

Fig. 5 showcases the detailed complexity results of the proposed aBsP detector, the RD-GAI-BP detector, and the BsP detector in $128 \times \{32, 16, 8\}$ MIMO scenarios with 256-QAM modulations. The related parameters in Table II are set as $K = 15$, $Q_L = 5$, and $n_m = 2$. For clarity and simplicity, the computational complexity is measured by the Fixed-Point Operations (FxPOs), which are computed by

$$\text{FxPOs} = k^2 \text{Mul.} + k \text{Add.} + m k^2 \text{Exp.}, \quad (15)$$

where k denotes the bit width of the Add. operations and m is the exponential bit width for the Exp. operations. Equation (15) originates from [19], in which the authors suppose that an addition with k bit width holds a computational

TABLE II
COMPUTATIONAL COMPLEXITY OF THE PROPOSED ABSP DETECTION

Algorithm	Mul.	Add.	Exp.
RD-GAI-BP [14]	$(4Q_L N_r N_t \sqrt{ \mathcal{A} }) + [4K N_t^2 + 2N_t \sqrt{ \mathcal{A} }]$	$(8Q_L N_r N_t \sqrt{ \mathcal{A} }) + [4K N_t^2 + 2N_t \sqrt{ \mathcal{A} }]$	$4Q_L N_r N_t \sqrt{ \mathcal{A} }$
BsP [§] [16]	$(4Q_L N_r N_t \sqrt{ \mathcal{A} }) + [4K N_t^2 + 2N_t \sqrt{ \mathcal{A} }]$	$(8Q_L N_r N_t \sqrt{ \mathcal{A} }) + [4K N_t^2 + 2N_t \sqrt{ \mathcal{A} }]$	0
Proposed aBsP	$(4Q_L N_r N_t n_m) + [4K N_t^2]$	$(8Q_L N_r N_t n_m) + [4K N_t^2 + 2N_t \sqrt{ \mathcal{A} }]$	0

[*]: Complexity of the initialization.

§: Configuration set $\mathcal{B}(1, 1)$ is utilized.

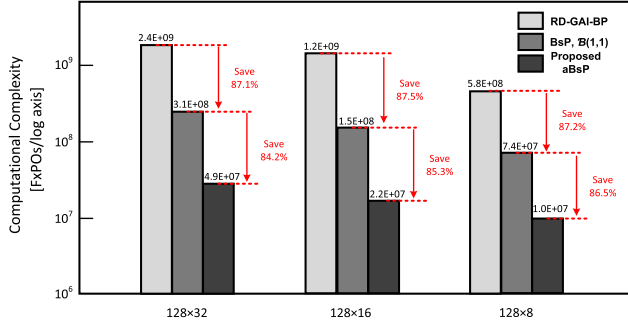


Fig. 5. Computational complexity results of the proposed aBsP detector, the RD-GAI-BP detector, and the BsP detector. The MIMO scenarios are set as $128 \times \{32, 16, 8\}$ antenna configurations with 256-QAM modulation.

complexity of k . Furthermore, according to [20], the computational complexity of a Mul. operation and an Exp. operation with k bit width is k^2 and mk^2 , respectively. According to the quantization scheme of the proposed aBsP detector (see Table III), here k and m are set as 14 and 8, respectively. From Fig. 5 it can be observed that the proposed aBsP detector achieves up to 86.5% complexity reduction benchmarking with the BsP with $\mathcal{B}(1, 1)$.

IV. MULTI-LEVEL AREA/POWER OPTIMIZATION TECHNOLOGY

In this section, we develop multi-level area/power optimization technology to further improve the hardware efficiency of the aBsP detector. Specifically, we utilize three main techniques: 1) *group-layered scheme with hyper-node configuration*, 2) *arithmetic approximate implementation*, and 3) *hybrid-precise quantization scheme* to improve the area efficiency and lower the power consumption. The details are provided as follows.

A. Group-Layered Scheme With Hyper-Node Configuration

The full-edge connected factor graph of the aBsP detector allows it to update the messages in parallel. However, this full-edge connected factor graph still involves bottlenecks for hardware implementation. On the one hand, the conventional flooding message-updating architecture suffers from low convergence speed, resulting in aggravated clock cycle (CC) consumption. On the other hand, the direct implementation of $2N_r$ and $2N_t$ nodes occupies too much area (e.g., for a 128×8 uplink massive MIMO system, there are 256 FN

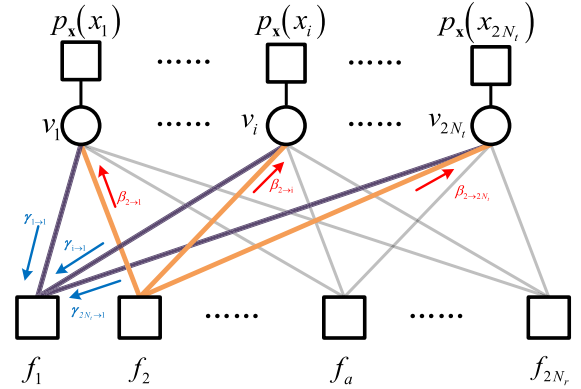


Fig. 6. Layered message-updating scheme of the aBsP detector.

nodes), leading to low hardware efficiency. To address this issue, we develop a group-layered message updating scheme with a hyper-node configuration technique here. The details are discussed as follows.

We draw inspiration from the BP decoding method of low-density parity-check (LDPC) codes in [21] and [22] and design a layered message-updating scheme for the aBsP detector. Benefiting from this layered scheme, the $2N_r$ FNs can be implemented with spatiotemporal folding/unfolding to significantly reduce the overall area. The layered message-updating scheme of the aBsP detector is shown as Fig. 6. However, due to the full-connected factor graph, each FN in this layered scheme is required to exchange messages with all $2N_r$ SNs in every iteration, resulting in high processing delay and intractable data write and read (W/R) control. To mitigate this problem, we develop a group-layered message-updating scheme with a hyper-node configuration technique here, as shown in Fig. 7. In particular, we evenly divide the $2N_r$ FNs into $2N_r/G$ groups. The G FNs in ξ -th group are merged together to build up a hyper-node, represented as g_ξ . With the node combination technique, the aBsP detector achieves $G \times$ CC reduction for each iteration, effectively enhancing the hardware efficiency.

It is noted that to better utilize the results of the wNSA-MMSE pre-processor, the aBsP detector in the first iteration employs a flooding message-updating scheme. The received β messages of each VN are accumulated and stored in the γ registers, which serve as the inputs for the next iteration. Fig. 8(a) reports the BER performance of the group-layered

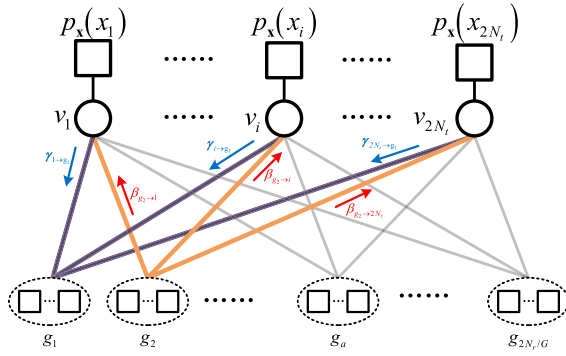


Fig. 7. Group-layered message-updating scheme with hyper-node configuration.

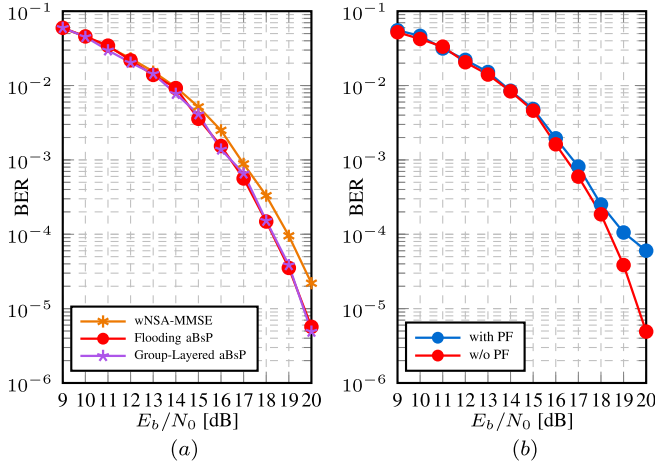


Fig. 8. BER performance comparisons in 128×8 MIMO scenario with 256-QAM modulation for the proposed aBsP detector with (a) different message-updating schemes, and (b) PF approximations.

aBsP with $G = 8$ in a 128×8 256-QAM MIMO system. The BER results demonstrate that the group-layered aBsP detector achieves consistent BER performance with the flooding aBsP detector. However, this group-layered scheme with the hyper-nodes sufficiently reduces the hardware resource consumption and processing delay. To improve the hardware efficiency, we utilize a pipeline architecture for the first message-updating iteration.

B. Approximate Arithmetic Circuits

Although the aBsP detector saves more than 87% of computational complexity compared with the original BsP detector, the direct implementation of the arithmetic operations still leads to long critical path delays, high processing latency, and inefficient power usage. Therefore, to achieve a lightweight computation, the combination law and tree-height reduction are exploited to shorten the wait time of each processing node while maintaining accuracy. In the wNSA-MMSE iteration, we first calculate the factor $\omega \mathbf{D}^{-1}$ and then combine the computation of Φ and $\mathbf{u}^{(1)}$ simultaneously, resulting in an overall reduction in processing time. Additionally, we benefit from the sparsity of diagonal matrices, which allows us to convert the matrix operations of the wNSA-MMSE into scalar operations. Alongside the aforementioned minor approximations, two main approaches are employed to enhance the

implementation efficiency of the Mul. and divisions, which are listed as follows.

1) *Approximations for divisions*: The division operations in the wNSA-MMSE pre-processor are implemented by piece-wise function (PF) approximations. For instance, the $2N_t$ divisions of $1/D_{ii}$, $i \in \{1, 2, \dots, 2N_t\}$, can be computed by performing the following PF approximation:

$$\hat{f}_{a,b}(1/D_{ii}, \xi) = \begin{cases} a_1 D_{ii} + b_1, & \text{if } D_{ii} < \xi, \\ a_2 D_{ii} + b_2, & \text{otherwise.} \end{cases} \quad (16)$$

Fig. 8(b) demonstrates that this PF approximation produces negligible performance loss within 0.2 dB at $\text{BER}=10^{-3}$, and suffers from a performance error floor below $\text{BER}=10^{-4}$. However, the PF approximation effectively shortens the critical path of the aBsP detector, which is now located on the delay of a single real-value multiply-accumulate (MAC) operation. It's worth mentioning that employing a more precise PF or other reciprocal functions like the Newton-Raphson function [23] can yield more accurate results. However, for the sake of simplicity, a simple PF with two segmentation is used in this work.

2) *Approximations for multiplications*: On the one hand, we re-designed all of the static multiplication operations in the aBsP detector by employing the well-known canonical signed digit (CSD) representation of integers. In particular, each multiplication involving the constellation points can be optimized by the CSD techniques, contributing to a higher implementation efficiency. More details of the CSD representation can be found in [24]. On the other hand, multiplications with fixed factors in this work are implemented following a low-cost logical expression by common subexpression elimination [25].

C. Hybrid-Precise Quantization Scheme

Following the conventional method, we acquire the integer and fractional bit-width for each variable by performing Monte Carlo simulations. However, to achieve a lower resource overhead and maintain comparable BER performance, we utilize a hybrid-precise quantization scheme for each variable, instead of the unified quantization. We define (q_s, q_i, q_f) as the bit-width representation of a variable, where q_s is the sign bit, q_i and q_f denote the bit-width of the integer and fraction, respectively. Take the quantization scheme of the wNSA-MMSE pre-processor as an example. In 128×8 256-QAM modulation MIMO scenarios, the quantization scheme of input variables in the matched filter and the Gram matrix is consistently set as (1, 11, 0). To achieve higher precision, the quantization scheme of the variables in the inverse matrix for the wNSA-MMSE pre-processor is confined to (1, 1, 12). The other variables in the wNSA-MMSE pre-processor utilize the quantization scheme as (1, 9, 1). More quantization details of the proposed aBsP detector are provided in Table III. The BER comparison results of the aBsP detector with this hybrid-precise fixed-point scheme and the float-point scheme are shown in Fig. 9. From Fig. 9 we can see that compared with the float-point aBsP detector, although the fixed-point aBsP detector suffers from larger performance loss with the

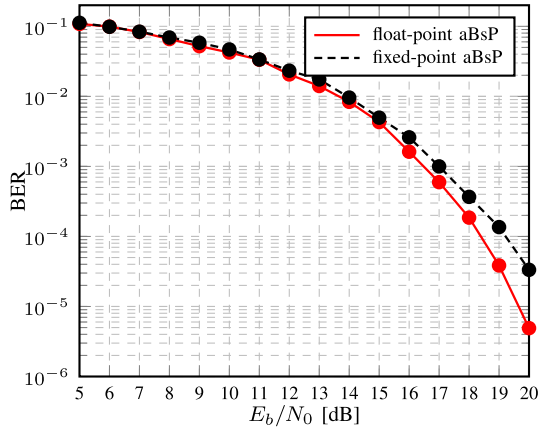


Fig. 9. The quantized BER of the 128×8 uplink MIMO detections with 256-QAM modulations in Rayleigh channel.

TABLE III
QUANTIZATION OF THE PROPOSED ABSPD

Variables	Quantization	Variables	Quantization
$\mathbf{H}$	(1, 3, 6)	$\mathbf{R}$	(1, 7, 2)
$\mathbf{D}$	(1, 9, 1)	$\tilde{\mathbf{y}}$	(1, 5, 4)
$\mathbf{W}$	(1, 9, 1)	$\tilde{\mathbf{y}}_2$	(1, 7, 3)
$\mathbf{A}$	(1, 9, 1)	$\tilde{\mathbf{y}}_t$	(1, 12, 1)
$\mathbf{D}^{-1}$	(1, 1, 12)	h_{ij}^2	(1, 4, 8)
Θ	(1, 9, 1)	$\mathbf{H}_u$	(1, 11, 2)
η	(0, 4, 0)	β	(1, 12, 1)
s_t	(1, 5, 0)	γ	(1, 7, 2)
R_{inc}	(1, 7, 2)		

increasing SNR, it achieves a comparable performance (within 0.5 dB loss) at the target BER of 10^{-3} . It is noted that for other modulations, the corresponding quantization scheme can be derived from the 256-QAM scheme by using linear compression.

V. ARCHITECTURE DESIGN AND ASIC IMPLEMENTATION OF THE ABSP DETECTOR

The details of the hardware architecture for the aBsP detector are introduced in the section. Furthermore, ASIC implementation results are provided to demonstrate the effectiveness of the aBsP detector.

A. Architecture Design

The entire architecture consists of two main arithmetic modules, namely the aBsP Message-updating PE and wNSA-MMSE PE, connected via the inter-data bus. Inter-iteration signals are stored in the corresponding registers. Control units are distributed in each sub-module to regulate the data flow following the corresponding message update.

1) *aBsP Message-updating PE*: Fig. 10(a) presents the architecture of message-updating PE for the proposed aBsP detector. Five main sub-blocks: LAT REC PROC X#0, EXT INFO PROC, LLH PR PROC 0#0, LLH PR PROC 1#0, and LLR INFO GEN #0 are involved in this PE. The LAT REC PROC X#0, $i \in \{1, 2, \dots, 2N_t\}$ is the lattice reduction processor, which is utilized to find the n_m closest lattice point according to the initialization information. More specific matching architecture is depicted in Fig. 11, where it

outputs the n_m closest matching points based on the related position of the estimated symbol vector. The f_{sw} signals in $2N_t$ Multiplexers (MUXs) decide whether to select the rough information $\hat{\mathbf{s}}_{\text{init}}$ from MMSE initialization or the output information $\mathbf{y}^{(l-1)}$ from the previous aBsP iteration for constellation matching. Here, the flag #Ci:PXQY in the i -th case means that the 1st and 2nd closest matching points are associated with the X-th and Y-th constellation points in the $\mathcal{A}$ -QAM modulation. The parallel $4N_t$ multiplexers in Fig. 11 are facilitated by the location of the constellations in the lattice reduction set and their corresponding information. This arrangement ensures the selection of potential combinations of channel matrix and transmits signals as $h_{ij}\sqrt{\mathcal{A}}(C_j)(n_m)$, where $j = \{1, 2, \dots, 2N_t\}$ and $i = \{1, 2, \dots, 2N_r/G\}$. The EXT INFO PROC denotes the extrinsic information processor which is utilized to compute the message R_{inc} . Details of this sub-block are illustrated in Fig. 10(b), in which the chosen combinations are stored in registers and accumulated using a 2-stage 16-input addition tree. The LLH PR PROC 0#0, and LLH PR PROC 1#0 are designed to compute the β messages for the first iteration and the remaining iterations, respectively. In particular, the restructuring of the architecture has allowed for the reorganization of the likelihood probability calculation $\beta(n_m) \in \mathbb{R}^{2N_t \times 2N_r}$ into a combination of parallel subtraction and parallel-vector multiplication, as depicted in Fig. 10(c). The LLR INFO GEN #0 represents the symbol LLR generator, which is utilized to compute the γ messages. More specifically, after computing the β messages, node combination is performed for different groups using $2N_t$ adder trees, as shown in Fig. 10(d). Here the signal f_{sw} decides that the inputs of the next iteration are acquired from the initial wNSA-MMSE or the preceding BsP iteration. The signal f_{y}^i determines whether to accumulate the β messages or to directly output the β messages. After the flooding message update in the 1st iteration, all of the γ messages in every group are summarized to activate the next iteration, and a simplified module LAT REC PROC 1#, is used to process the truncated lattice points for the remaining iterations.

2) *wNSA-MMSE PE*: Based on the pre-computed matching filtered vector $\mathbf{y} \in \mathbb{R}^{2N_t \times 1}$ and Gram matrix $\mathbf{A} \in \mathbb{R}^{2N_t \times 2N_t}$, the overall structure of wNSA-MMSE PE shown as Fig. 12 generates the initial information $\mathbf{s}_{\text{init},i}$, $i = \{1, 2, \dots, 2N_t\}$ for the following aBsP iterations. As discussed in Section IV, the PF approximation-based matrix-inversion, denoted as

$$\omega \mathbf{D}^{-1} = \frac{\omega}{\text{diag}(\mathbf{A}) + \text{diag}(\Lambda)} = \hat{f}_{\mathbf{a},\mathbf{b}}(\omega \mathbf{D}^{-1}, \xi), \quad (17)$$

is computed by the GRAM INV #i module with $4N_t$ MACs in 2 cycles, where $(a_1 = -0.1992, b_1 = 0.914)$ and $(a_2 = -0.4375, b_2 = 1.3125)$. Subsequently, we employ parallel-vector multiplication groups for $\mathbf{u}^{(1)}$ (using $2N_t$ MACs) and Φ (requiring $2N_t \times 2N_t$ MACs). A similar multiplication group is adopted to compute the primary iteration of wNSA-MMSE. Moreover, We employ $2N_t$ two-stage adder trees with 17-input to collect data, enabling high-throughput calculations between matrices and vectors. Meanwhile, $2N_t$ f_L signals control the iteration process and determine whether outputs are exported

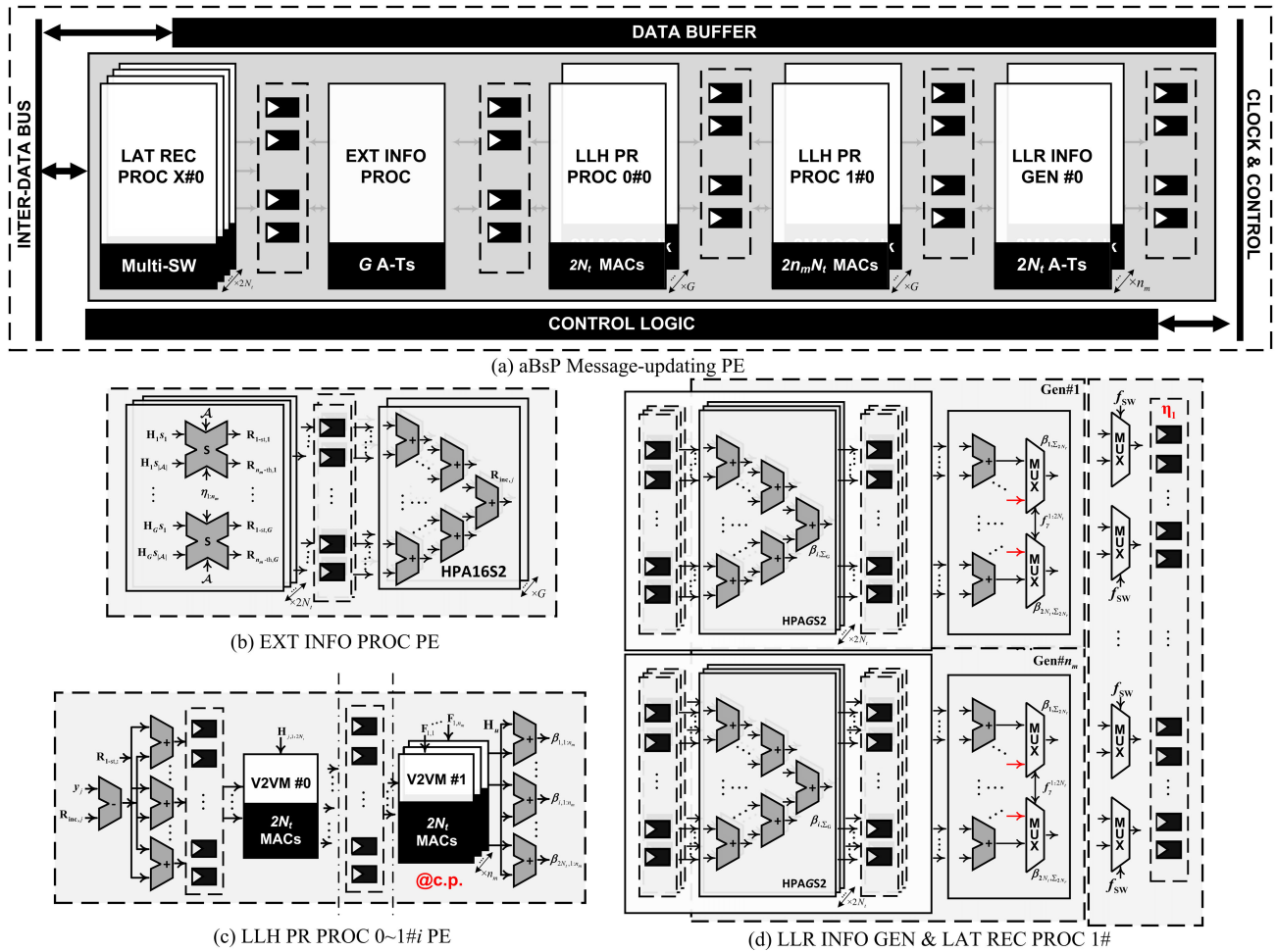


Fig. 10. Hardware architecture of aBsP Message-passing PE in 128×8 MIMO with 256-QAM modulation; LAT REC PROC 0/1#: lattice reduction processor for the i -th user in 0/1-mode; EXT INFO PROC: extrinsic information processor; LLH PR PROC 0/1#: likelihood probability processor in 0/1-mode; LLR INFO GEN: log-likelihood ratio generator; HPArSy: x -input hybrid precision addition with y -stage pipeline. Multi-SW, A-T, and MAC denote *multi-road switch*, *adder-tree*, and *multiply-accumulator*, respectively.

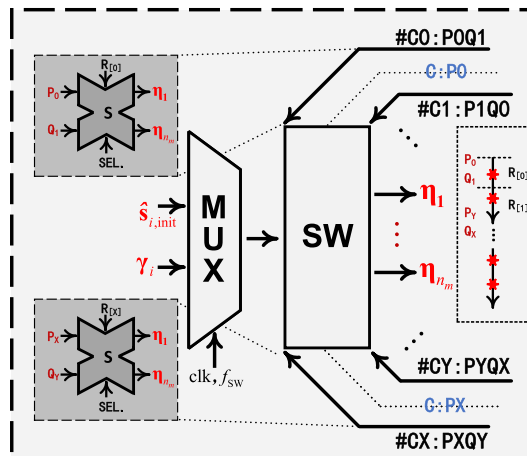


Fig. 11. Hardware architecture for LAT REC PROC 0# with n_m surviving constellations in $\mathcal{A}$ -QAM.

to the subsequent module or back to the beginning of NSA CVR PE module, as depicted in Fig. 12.

B. Implementation Results

A 128×8 256-QAM aBsP detector is implemented in SMIC 40 nm LL 1P9M CMOS technology, which is synthesized on

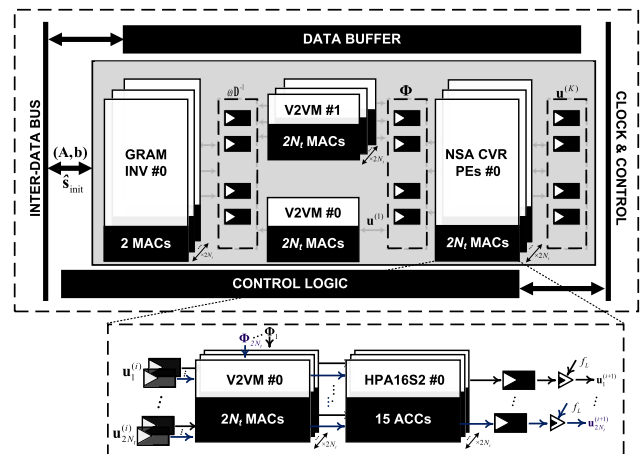


Fig. 12. Hardware architecture for wNSA-MMSE PE in A-QAM; GRAM INV: Gram matrix inversion; NSA CVR PE: NSA convergence PE; V2VM: vector-to-vector multiplication; ACC: Accumulator.

Synopsys® Design Compile, placed and routed via Synopsys® IC Compiler. The annotated toggle rate of the gate-level netlist is converted into switching activity interchange format (SAIF) for Prime-Time PX to measure the time-based chip-only power dissipation. The aBsP detector achieves the throughput

TABLE IV
ASIC IMPLEMENTATION RESULTS & COMPARISON OF STATE-OF-THE-ART MASSIVE MIMO DETECTORS

Detector	This work	Peng [3] TCAS-I'18	Tang [6] ISSCC'18	Jeon [7] SSCC-L'18	Tan [8] TVT'20	Tan [9] TCAS-II'20	Chen [11] TVLSI'23
Algorithm	aBSP	WeJiMMSE	EPD	LAMA	BP-DNN	EP-INSAs	EPA-SORI
Modulation	256-QAM	64-QAM	256-QAM	256-QAM	16-QAM	256-QAM	256-QAM
$N_t \times N_r$	128×8	128×8	128×16	128×32	128×8	128×8	128×32
Technology [nm] (@Vol.)	40(@1.1 V)	65(@1.0 V)	28(@1.0 V)	28(@0.9 V)	65(@1.0 V)	65(@1.0 V)	28(@0.9 V)
Gate Count [kGE]	286	1070	3607	1036	1478	1243	3657
Core Area [mm ²]	0.68	2.57	2	0.37	0.8	3.68	1.38
Frequency [MHz]	1334.53	680	569	400	340	500	1087
Power [mW]	437.5	650	127	151	225.9	573.2	225.5
Scaled T/P ^① [Mbps]	790.52	589.33	180.15	248.03	1000.00	294.67	3,043.56
Scaled EE ^② [Mbps/mW]	1.81	1.22	0.82	0.77	2.34	1.75	6.33
Scaled GE ^③ [Mbps/kGE]	2.77	0.55	0.05	0.24	0.68	0.24	0.83
Scaled AE ^④ [Mbps/mm ²]	1,162.53	605.53	44.14	328.47	717.56	972.63	1,081.20

^① Scaled to $N_t = 8$, 256-QAM and SMIC 40 nm with 1.1 V supply. ^② EE represents the energy efficiency as $\frac{T/P}{\text{Energy}}$.

^③ GE represents the gate efficiency as $\frac{T/P}{\text{GateCount}}$. ^④ AE represents the area efficiency as $\frac{T/P}{\text{CoreArea}}$.

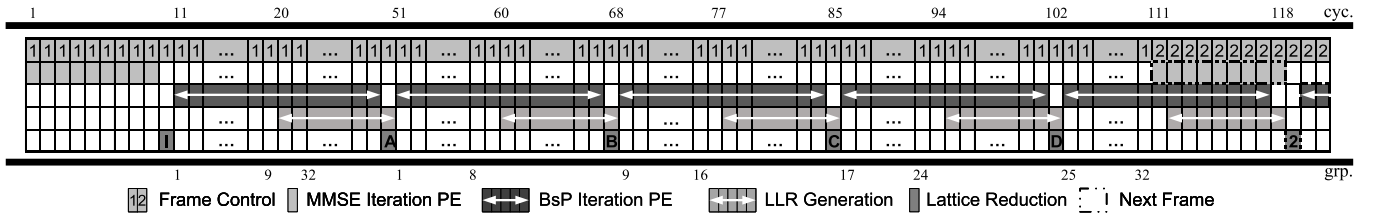


Fig. 13. Time schedule for aBSP detector in 128×8 MIMO with 256-QAM.

formulated as

$$T/P = \frac{\log_2(|\mathcal{A}|) \times N_t}{T_{\text{BsP, FL}} + T_{\text{BsP, LAY}}} \times f_{\text{clk}}, \quad (18)$$

where f_{clk} represents the clock frequency, and the critical path of aBSP detector is observed in the scalar multiplication employing a Booth-recoded Wallace-tree model. Fig. 13 shows the number of CCs spent for the flooding and group-layered message updates in each iteration. It can be seen that the CC consumption of the flooding and group-layered scheme are $T_{\text{aBSP, FL}} = 40$ and $T_{\text{aBSP, LAY}} = 68$, respectively.

Table IV shows the implementation comparison results of the proposed aBSP detector and the other massive MIMO detectors. For a fair comparison, all of the detectors are normalized to the same feature size and threshold voltage with

$$f_{\text{clk}} \propto \zeta, \text{Area} \propto \frac{1}{\zeta^2}, \text{Energy} \propto \frac{v^2}{\zeta}, \quad (19)$$

where ζ and v denote the scaling factor and voltage factor [3]. In addition, hardware metrics are further normalized for a clearer comparison, considering the same channel configuration ($N_t = 8$) and modulation scheme (256-QAM). Hence, throughput, chip area, and power consumption are all scaled by $\frac{8 \log_2 N_t}{N_t \log_2 8}$ [26]. Meanwhile, the area normalization for the modulation is determined by the computational complexity for soft-output detectors in [27]. Several observations can be drawn from Table IV. First, the proposed aBSP detector enjoys the highest frequency (1.334 GHz), benefiting from its pipeline and critical path optimization based on the algorithm approximations and hardware design. Second, the proposed aBSP detector achieves $1.92\times$ and $5.04\times$ scaled AE and GE

TABLE V
ASIC IMPLEMENTATION DETAILS

Technology	SMIC 1P9M CMOS
Process Corner	Typical Typical
PVT Variations	(40 nm, 1.1 V, 25 °C)
Chip Size/Utilization	(1,157.15 mm ² , 50.69%)
Gate Count@(SNR, BER)	286 kGE @ (16.5dB, 10^{-3})
Scaled AE@(SNR, BER)	1,162.53 Mbps/mm ² @ (16.5dB, 10^{-3})
Scaled EE@(SNR, BER)	1.81 Mbps/mW @ (16.5dB, 10^{-3})

compared with the WeJiMMSE detector [3], respectively. It is noted that the proposed aBSP detector also enjoys better BER performance than the LMMSE detector, which is demonstrated in Figs. 2-3. Third, compared with the 28-nm LAMA [7] detector, the proposed aBSP detector shows $11.54\times$ GE, $3.54\times$ AE, and $11.54\times$ EE, respectively. Compared with the BP-DNN detector [8], the proposed aBSP detector exhibits $4.07\times$ GE and $1.62\times$ AE improvements. Finally, benchmarking with the recent art (EPA-SORI) [11], the proposed aBSP detector succeeds in achieving $3.34\times$ GE and $1.08\times$ AE. Here we note that although the EPA-SORI detector represents comparable hardware efficiency with the proposed aBSP detector, it suffers from aggravated performance error floor in massive MIMO systems with challenging antenna configurations and high-order modulations.

Table V further details the implementation results of the proposed aBSP detector. It can be seen that the overall detector involves 286k logic gates, and occupies a core area of 0.682 mm². The chip achieves a 790.52 Mb/s throughput at $f_{\text{clk}} = 1.334$ GHz with a 437.5 mW power dissipation

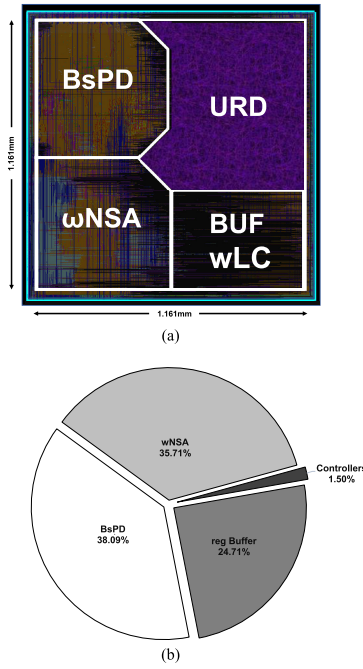


Fig. 14. (a) Layout photo and (b) area breakdown of the proposed aBsP detector core; URD includes other baseband processing modules with a proportion of 49.31%, which is not related to the proposed aBsP detector; Two main modules: wNSA and BsPD occupy 35.71% and 38.09% core area, respectively.

at the target BER of 10^{-3} . Fig. 14 illustrates the layout photo and the area breakdown of the proposed aBsP detector. Two main modules: wNSA and BsPD occupy 35.71% and 38.09% core area, respectively. The results showcase the integration of high-density vectorized computations enabled by the co-optimization technology in the aBsP and wNSA-MMSE modules.

VI. CONCLUSION

The first hardware implementation of the BsP detector is proposed. Two key features: approximate initialization and simplified message updates are utilized to lower the costs (more than 84%) with acceptable performance penalization. Multi-level area/power optimizations: group-layered scheme, arithmetic approximate implementation, and hybrid-precise quantization are introduced. As a proof of a concept, a 0.68 mm^2 790.52 MHz 437.5 mW aBsP ASIC is implemented in 40 nm CMOS for 128×8 256-QAM massive MIMO. Compared with the recent arts, the aBsP detector achieves $3.34 \times$ GE and $1.08 \times$ AE, demonstrating the feasibility and advantages of the BsP. Future work will be directed towards the coded BsP, and the aBsP-based all-message-passing unified baseband processor.

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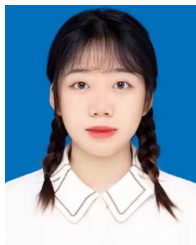
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